STYLIANOS GIAGKOS

Data & AI Consultant | Data Scientist | Electrical & Computer Engineer

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PROFILE

Data & AI Consultant at EY with hands-on experience delivering cloud data pipelines and AI-driven analytics for large-scale digital transformation programs across energy and sustainability sectors. Combining an MEng in Electrical & Computer Engineering with an MSc in Data Science (AUEB), I specialize in end-to-end data engineering on GCP, applied machine learning, and deep learning research - from building production Medallion architectures in BigQuery to developing clinical audio classification models with PyTorch. Currently contributing to AI-powered operational analytics for a leading oil & energy organization in Saudi Arabia.

WORK EXPERIENCE

Data & AI Consultant Jul 2025 – Present

EY Greece | AI & Data Department

- Architected and maintained data pipelines on **GCP** (BigQuery, Dataform, Cloud Storage) following **Medallion architecture** for a large-scale digital transformation program serving a leading oil & energy organization in Saudi Arabia.
- Proposed **forecasting models**, **anomaly detection**, and scenario analytics that directly support operational efficiency and sustainability decision-making.
- Built a custom **Bash-based orchestration layer** automating ingestion workflows across GCP and PostgreSQL, including Dataform executions, GCP operations, and structured run logging.
- Designed reusable SQL logic, **SCD Type 2** tracking for historical data, and rule-based data quality validations with schema checks and exception handling.
- Collaborated across multi-regional engineering and sustainability teams; contributed to documentation and stakeholder-facing deliverables.

QC Electrical Engineer Dec 2023 – May 2024

Sunlight Group | Xanthi, Greece

- Defined QC checkpoints and compliance documentation for the **HVESS assembly line**, tracking production issues and coordinating timely claim resolution across EU and US markets.
- Conducted **Root Cause Analysis (RCA)**; monitored electronics and mechanical design findings on the production line; delivered **data-driven reports** with key metrics and operational insights to upper management.
- Supported prototyping assessment and supply agreement reviews from a QC/QA perspective.

Data & Analytics Intern Jul 2022 – Aug 2022

EY Greece | Data & Analytics

- Contributed to an **MMX Modeling project** for an international pharmaceutical company, developing Python and SQL scripts for analytical tasks.
- Participated in client meetings and sprint workflows, gaining hands-on experience in end-to-end data-driven consulting.

SELECTED PROJECTS

Explainable AI for Real Estate Investment Valuation MSc Thesis

Python, XGBoost, Random Forest, SHAP (TreeSHAP), Optuna, Spatial Cross-Validation

- Developed an Explainable Automated Valuation Model (AVM) for predicting residential sale and rental prices in the Greek real estate market using ~18,000 listings. Implemented log-transformed pricing models with Bayesian hyperparameter optimization (Optuna) and Spatial Cross-Validation (GroupKFold). Built an explainability framework using SHAP waterfall plots and counterfactual analysis for transparent investment decision support. Grade: 9/10

Deep Learning for Respiratory Disease Classification MSc Research

PyTorch, Librosa, ResNet50, DenseNet121, Domain Adaptation (GRL)

- Built a clinical audio classification pipeline for 10-class respiratory disease diagnosis using CNN architectures and transfer learning. Applied spectral preprocessing and domain adaptation via Gradient Reversal Layer. Grade: 9.5/10

SemEval 2025: Food Hazard Detection Challenge NLP / Multi-label

HuggingFace Transformers, BERT, RoBERTa, LightGBM, ResNet101

- Developed NLP models for multi-label food hazard classification combining transformer fine-tuning with image clustering using deep learning embeddings. Grade: 10/10

Real-Time Streaming Pipeline with Kafka & PySpark Big Data Eng.

Apache Kafka, PySpark Structured Streaming, Cassandra, Hadoop HDFS

- Designed distributed real-time data pipelines with structured streaming, stateful transformations, and streaming joins on Hadoop ecosystem. Grade: 10/10

E-Commerce Product Classification with Graph Mining Data Science

NetworkX, BERT, TF-IDF, LightGBM, Scikit-learn

- Built an ensemble model combining text features (TF-IDF, BERT embeddings) with graph-based features for multi-label product categorization. Grade: 10/10

Pretrained BERT Models in Gene–Metabolite Correlation MEng Thesis

Python, BERT, BioBERT, PubMedBERT, Biomedical Text Mining

- Benchmarked domain-specific BERT variants for extracting gene-metabolite relationships from medical literature using NER and relation extraction.

EDUCATION

MSc in Data Science Sep 2024 – Oct 2025

Athens University of Economics and Business (AUEB)

Thesis: Automated Real Estate Investment Value Assessment System Using AI (9/10) | GPA: 8.73/10

Coursework: Machine Learning, Deep Learning, Text Analytics, Numerical Optimization, Large-Scale Data Management, Data Visualization

Key grades: Practical Data Science 10/10, Large Scale Data Management 10/10, Data Science Challenge 10/10

MEng in Electrical & Computer Engineering Oct 2018 – Dec 2023

Democritus University of Thrace

Thesis: Comparative Study of Pretrained BERT Models in Genes–Metabolites Correlation in Medical Literature | GPA: 7.79/10

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript (SQLX/Dataform), R, C++, HTML/CSS
ML / Deep Learning	PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LightGBM, NLP (BERT, RoBERTa, GloVe)
Cloud & Data Eng.	GCP (BigQuery, Cloud Storage, AlloyDB, Dataform), ETL/ELT Pipelines, Medallion Architecture, PostgreSQL
Big Data	Apache Kafka, PySpark, Hadoop (HDFS, MapReduce), Cassandra
Visualization / BI	Tableau, Matplotlib, Seaborn, Plotly, ggplot2, Pandas
Tools & DevOps	Git, Docker, Linux, Bash Scripting, API Integration

CERTIFICATIONS

- Neural Networks and Deep Learning - DeepLearning.AI, Coursera (2021)
- Python for Data Science and Machine Learning Bootcamp - Udemy (2022)
- PMI Project Management Professional (PMP) - In progress

LANGUAGES

Greek (Native) **English** (Professional proficiency) **German** (Elementary proficiency)